

Chapter 2

HTML Part 5

Forms

INT150 Web Technologies (2/2025)

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Topics

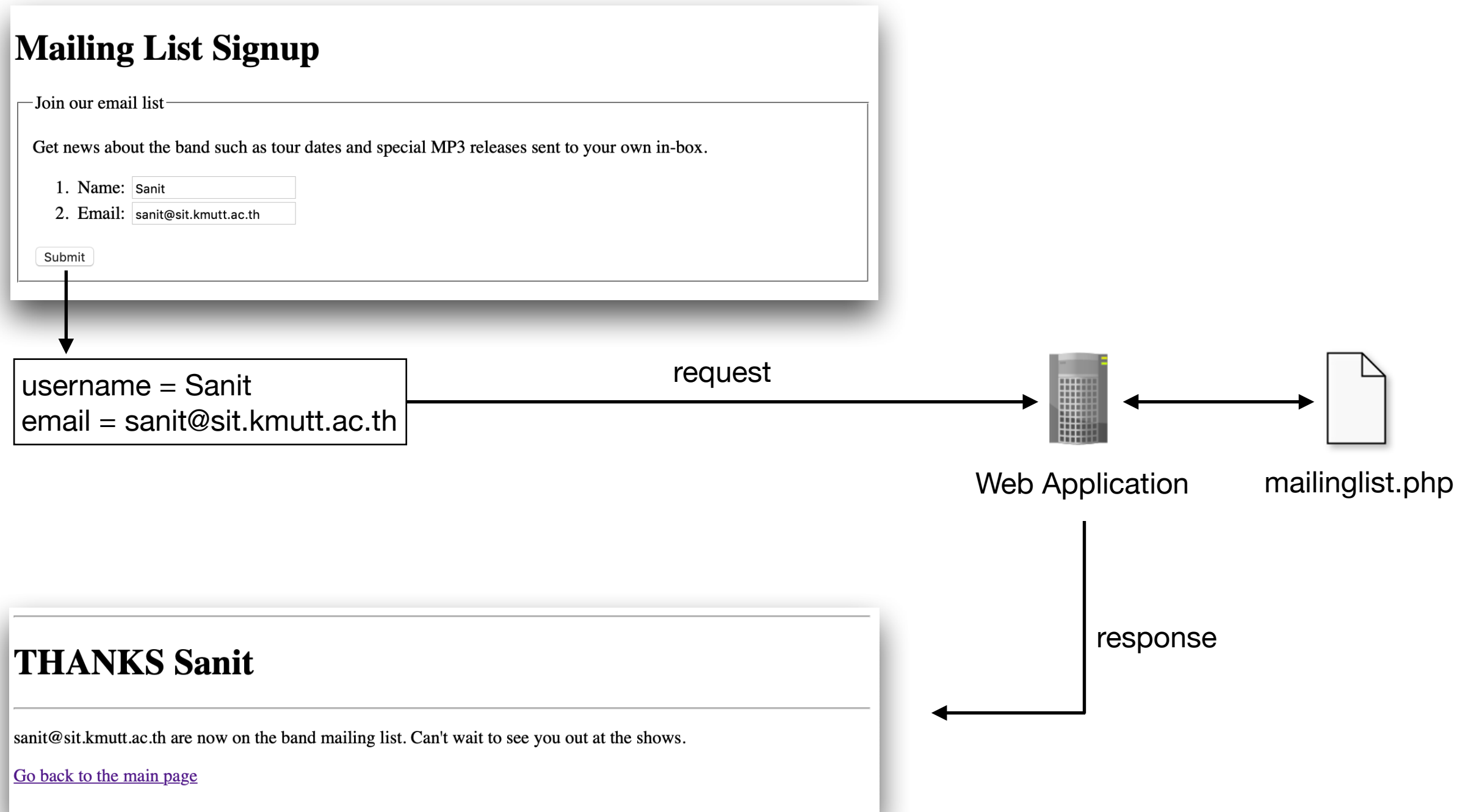
- What will you learn after finishing this class:
 - HTML Forms and form attributes
 - Form Controls
 - Text entry controls
 - Password field
 - Radio and Checkbox buttons
 - Drop-down and Scrolling menus
 - Submit and reset buttons
 - Hidden controls
 - Fieldset and Legend element

HTML Forms

- HTML **Forms** are used to collect some information from the site visitors.
 - Ex. During user registration, user information such as name, email address, credit card are required.
- A **form** will take input from the site visitor and then will **post** it to a back-end application such as **CGI**, **ASP** Script or **PHP** script.
- The **back-end application** processes the information collected by the form and returns the appropriate **response** to the site visitor.

From data entry to response

signup.html



signup.html

```
<!DOCTYPE html>
<html>
  <head>
    <title>Mailing List Signup</title>
    <meta charset="utf-8">
  </head>
  <body>
    <h1>Mailing List Signup</h1>
    <form action="mailinglist.php" method="post">
      <fieldset>
        <legend>Join our email list</legend>
        <p>Get news about the band such as tour dates and special MP3
        releases sent to your own in-box.</p>
        <ol>
          <li>
            <label for="firstlast">Name:</label>
            <input type="text" name="username" id="firstlast">
          </li>
          <li>
            <label for="email">Email:</label>
            <input type="text" name="email" id="email">
          </li>
        </ol>
        <input type="submit" value="Submit">
      </fieldset>
    </form>
  </body>
</html>
```

Mailing List Signup

Join our email list

Get news about the band such as tour dates and special MP3 releases sent to your own in-box.

1. Name:

2. Email:

Demo

mailinglist.php

```
<!DOCTYPE html>
<html>
  <head>
    <meta charset="utf-8" />
    <title>Mailing List Signup Result</title>
  </head>
  <body>
    <hr>
    <h1>THANKS <?php echo $_POST["username"]; ?></h1>
    <hr>
    <p><?php echo $_POST["email"]; ?> are now on the band mailing list.
    Can't wait to see you out at the shows.</p>
    <a href="signup.html">Go back to the main page</a>
  </body>
</html>
```

THANKS Sanit

sanit@sit.kmutt.ac.th are now on the band mailing list. Can't wait to see you out at the shows.

[Go back to the main page](#)

<form> element

- **Forms** are added to the web pages using the `<form>` element.
- The `<form>` element is **a container** for all the content of the form such as input field, buttons and drop-down menu (collectively known as **form controls**).
- The `<form>` element may contain block elements such as **p**, **h1**, and **lists**. However, it may not contain another form element.

Form Attributes

Attribute	Description
action	The location (URL) of the application or script (sometimes called the action page) that will be used to process the form
method	The method to be used to upload data to the server: POST or GET

```
<form action="mailinglist.php" method="post">
```


Scripting Language

- PHP (.php)
- Microsoft's ASP.NET (.asp)
- Ruby on Rails
- JavaServer Pages (.jsp)
- Python
- NodeJS (.js)

method attribute

- The `method` attribute specifies how the information should be sent to the sever.
- Data gathered from the form
`username = Sanit`
`email = sanit@sit.kmutt.ac.th`
- Browser encodes that information for its trip to the server
`username=Sanit&email=sanit%40sit.kmutt.ac.th`
- Two methods for sending this encoded data to the server:
 - POST
 - GET (`Default`)

POST method

- The browser sends a server request containing some special headers followed by the data.
- Only the **server** sees the content of this request.
- The **POST** method is the best method for sending **secure information** such as credit card or other **personal information**.
- The **POST** method has **no size limitations** and can be used to **send a lot of data**.
- Form submissions with the **POST** method **cannot** be **bookmarked**.
- The **POST** method is used when the form is used to upload a file.

GET method

- When the **GET** method is used, the submitted form data will be visible in the page address bar. The question mark character separates the URL from the following data:

<https://int102.sit.kmutt.ac.th/webtech140/maillinglist.php?username=Sanit&email=sanit%40sit.kmutt.ac.th>

- The length of a URL is **limited** (about 3,000 characters)
- The **GET** method allows users to be able to bookmark the results of a form submission.
- The **GET** method is not be used for the forms with private personal or financial information.

Name Attribute

- The **name** attribute provides the **variable name** for the **control**.
- All form control elements must include a **name** attribute so the form-processing application can get the information.
- Each **variable** in a **form** must be named uniquely.

```
<h1>Mailing List Signup</h1>
<form action="mailinglist.php" method="post">
  <fieldset>
    <legend>Join our email list</legend>
    <p>Get news about the band such as tour dates and special MP3
    releases sent to your own in-box.</p>
    <ol>
    <li>
      <label for="firstlast">Name:</label>
      <input type="text" name="username" id="firstlast">
    </li>
    <li>
      <label for="email">Email:</label>
      <input type="text" name="email" id="email">
    </li>
    </ol>
    <input type="submit" value="Submit">
  </fieldset>
</form>
```

signup.html

```
<hr>
<h1>THANKS <?php echo $_POST["username"]; ?></h1>
<hr>
<p><?php echo $_POST["email"]; ?> are now on the band mailing list.
Can't wait to see you out at the shows.</p>
```

mailinglist.php

Form Controls

- There are different types of form controls that can be used to collect data using HTML form.
 - Text entry controls
 - Specialized text entry controls
 - Radio buttons
 - Checkbox buttons
 - Pull-down and scrolling menus
 - File selection and upload control
 - Hidden controls
 - Clickable buttons
 - Submit and reset buttons
 - Dates and times ([HTML5](#))
 - Numerical controls ([HTML5](#))
 - Color picker control ([HTML5](#))

<input> element

- The <input> element can be displayed in several ways, depending on the type attribute.

<input type = "text">

text

<input type = "number">

Number

<input type = "radio">

☐ radio

<input type = "checkbox">

☐ check box

<input type = "submit">

Single-line text field

`<input type = "text">`

- is used for entering a **single word** or line of **text**.
- It is the **default type** (when omitting the type attribute).
- Attributes
 - **name** defines the variable name
 - **value** specifies default text when the form is loaded.
 - **size** specifies the width of the text input control (characters)
 - **maxlength** specifies the maximum character limit.

<https://stackoverflow.com/questions/25247565/difference-between-maxlength-size-attribute-in-html>

Text Field

```
<form >  
  First name: <input type = "text" name = "first_name" >  
  <br>  
  Last name: <input type = "text" name = "last_name" >  
</form>
```

First name:

Last name:

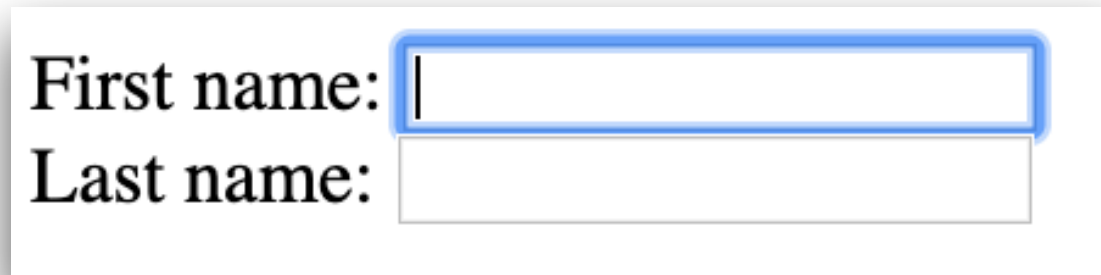
Demo

<label> element

- The <label> element defines a label for a <input>, <textarea>, <select> or <button> element.
- The <label> does not render as anything special for the user. It provides a usability improvement for mouse users.
- When the user clicks on the text within the <label> element, it toggles the control.
- The for attribute of the <label> element should be equal to the id attribute of the related element to bind them together.

Text Field with Label

```
<form >
  <label for = "first_name">First name: </label>
  <input type = "text" name = "first_name" id="first_name" >
  <br>
  <label for = "last_name">Last name: </label>
  <input type = "text" name = "last_name" id="last_name" >
</form>
```



Demo

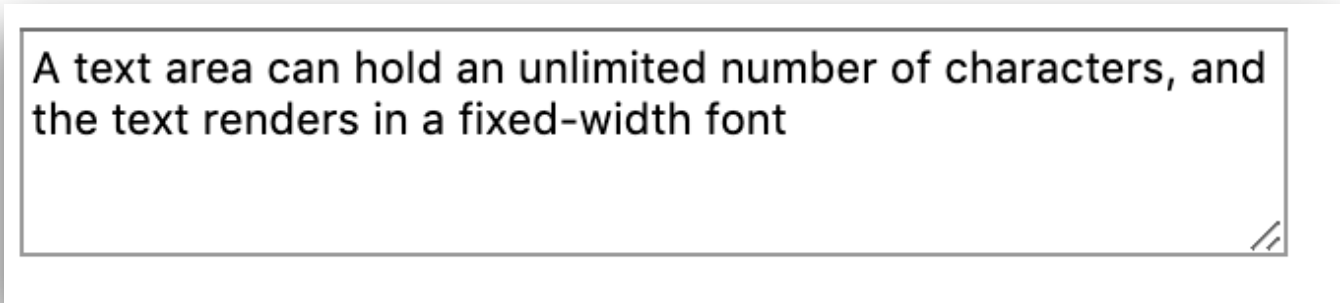
Multiline text entry field

`<textarea>...</textarea>`

- This is used when the user is required to give details that may be longer than one line of text.
- Attributes
 - `name`
 - `rows` specifies the number of rows of text area box
 - `cols` specifies the number of columns of text area box
 - `placeholder` (HTML5) provides a short hint of how to fill the field.
- The `rows` and `cols` attributes are a way of specifying the size of the text area.
- Unlike the empty `input` element, the `content` can be put between the opening and closing tags in the `textarea` element.

<textarea> element

```
<form >  
  <textarea rows="4" cols="50"> A text area can hold an unlimited nu  
mber of characters, and the text renders in a fixed-width font  
  </textarea>  
</form>
```



A text area can hold an unlimited number of characters, and
the text renders in a fixed-width font

Demo

Specialized text entry fields

- Specialized text entry fields

- Password entry field

`<input type = "password">`

- Search field ([HTML5](#))

`<input type = "search">`

- Email address ([HTML5](#))

`<input type = "email">`

- Telephone number ([HTML5](#))

`<input type = "tel">`

- Location (URL) ([HTML5](#))

`<input type = "url">`

- Range ([HTML5](#))

`<input type = "range">`

https://www.w3schools.com/tags/tag_input.asp

Password entry field

`<input type = "password">`

- A **password** field works like a text entry field, except the characters are obscured from view using **asterisk** (*) or **bullet** (•) or another character determined by the browser.
- The form does **not encrypt** the information, so it should not be considered a real security measure.

Password field

```
<form>  
  <label for="form-pswd">Password</label>  
  <input type="password" name="pswd" maxlength="8" id="form-pswd">  
</form>
```



Password

Demo

Range type (slider control)



- Range type creates a slider control for selecting a numeric value within a specific range.
- The exact value is not generally considered important when using this input type.
- It is commonly used for settings like volume, brightness or game difficulty where an approximate value is sufficient.
- Attributes:
 - min specifies the minimum value allowed. The default is 0.
 - max specifies the maximum value allowed. The default is 100.
 - step defines the legal number intervals (increments) for the slider.
 - value sets the initial value of the slider.

Form Control Attributes

- **Data Validation**

- required -- indicates whether this element is required to fill out or not.
- readonly -- indicates whether the element can be edited.
- disabled -- indicates whether the user can interact with the element.
- pattern -- defines a regular expression which the element's value will be validated against.

- **User Experience**

- placeholder -- provides a hint to the user of what can be entered in the field.
- autofocus -- indicates whether the element should be focused on page load.

- **Data control**

- name -- name of element used by the server to identify the fields in form submits.
- value -- defines a default value which will be displayed in the element on page load.
- list -- identifies a list of pre-defined options to suggest to the user.

Radio and Checkbox buttons

- Both **radio** and **checkbox buttons** make it simple for visitors to choose from a number of provided **options**.
- **Radio buttons** are used when out of many options, just **one option** is required to be selected.
- **Checkboxes** are used when **more than one option** is required to be selected.

Radio buttons

```
<input type = "radio" name = "variable" value = "value">
```

- The **name** attribute is required and play an important role in binding multiple radio inputs into set.
- When a number of radio button inputs are provided with the **same name value**, they create **a group of mutually exclusive options**.

Radio buttons

```
<form>
  <p>Please select your preferred contact method:</p>
  <div>
    <input type="radio" id="contactChoice1"
      name="contact" value="email">
    <label for="contactChoice1">Email</label>

    <input type="radio" id="contactChoice2"
      name="contact" value="phone">
    <label for="contactChoice2">Phone</label>

    <input type="radio" id="contactChoice3"
      name="contact" value="mail">
    <label for="contactChoice3">Mail</label>
  </div>
  <div>
    <button type="submit">Submit</button>
  </div>
</form>
```

Please select your preferred contact method:

☐ Email ☐ Phone ☐ Mail

Demo

Checkbox buttons

`<input type = "checkbox">`

- As with radio buttons, **creating groups** of checkboxes is by assigning them the **same name value**.
- When more than one checkbox may be checked at a time, the value of every checked button will be sent to the server when the form is submitted.
- Checkboxes don't necessarily need to be used in groups.

Checkbox buttons

```
<form>
  <p>Please select your preferred contact method:</p>
  <div>
    <input type="checkbox" id="contactChoice1"
      name="contact[]" value="email">
    <label for="contactChoice1">Email</label>

    <input type="checkbox" id="contactChoice2"
      name="contact[]" value="phone">
    <label for="contactChoice2">Phone</label>

    <input type="checkbox" id="contactChoice3"
      name="contact[]" value="mail">
    <label for="contactChoice3">Mail</label>
  </div>
  <div>
    <button type="submit">Submit</button>
  </div>
</form>
```

Please select your preferred contact method:

☐ Email ☐ Phone ☐ Mail

Demo

Menus

```
<select name = "variable">  
  <option> choice 1 </option>  
  <option> choice 2 </option>  
  <option> choice 3 </option>  
</select>
```

- Menus provide **a list of choices** by putting them in a drop-down or scrolling menu.
- By default, the **content of the chosen option** is what gets passed to the web application when the form is submitted.
- To send **a different value** than what appears in the menu, use the **value** attribute to provide an overriding value.

```
<select name = "variable">  
  <option value = "1"> choice 1 </option>  
  <option value = "2"> choice 2 </option>  
  <option value = "3"> choice 3 </option>  
</select>
```


Drop-down menus

- The `select` element displays as a drop-down menu (also called a pull-down menu) by default when no size is specified or if the `size` attribute is set to 1.
- Only one item can be selected.

Drop-down menus

```
<form>
  <label for="pet-select">Choose a pet:</label>

  <select id="pet-select" name="pet-select">
    <option value="">--Please choose an option--</option>
    <option value="dog">Dog</option>
    <option value="cat">Cat</option>
    <option value="hamster">Hamster</option>
    <option value="parrot">Parrot</option>
    <option value="spider">Spider</option>
    <option value="goldfish">Goldfish</option>
  </select>
</form>
```

Choose a pet: --Please choose an option--

Choose a pet ✓ --Please choose an option--

Dog

Cat

Hamster

Parrot

Spider

Goldfish

Demo

Scrolling menus

- To make the menu display as a scrolling list
 - specify the number of lines to be visible using the size attribute (`size >1`).
- The `multiple` attribute allows users to make more than one selection from the scrolling list.
- When the browser detects the `multiple` attribute, it displays a small scrolling menu automatically by default.
- Use the `selected` attribute in an option element to make it the default value for the menu control.

Scrolling menus

```
<form>
  <label for="pet-select">Choose a pet:</label>

  <select id="pet-select" name="pet-select" size="6">
    <option value="dog">Dog</option>
    <option value="cat">Cat</option>
    <option value="hamster">Hamster</option>
    <option value="parrot">Parrot</option>
    <option value="spider">Spider</option>
    <option value="goldfish">Goldfish</option>
  </select>
</form>
```

Choose a pet:

Dog

Cat

Hamster

Parrot

Spider

Goldfish

Demo

Scrolling menus with default value

```
<form>
  <label for="pet-select">Choose a pet:</label>

  <select id="pet-select" name="pet-select" size="6">
    <option value="dog">Dog</option>
    <option value="cat" selected>Cat</option>
    <option value="hamster">Hamster</option>
    <option value="parrot">Parrot</option>
    <option value="spider">Spider</option>
    <option value="goldfish">Goldfish</option>
  </select>
</form>
```

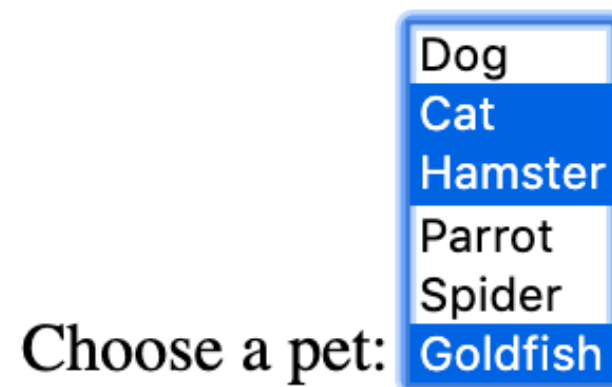
Choose a pet:

Dog
Cat
Hamster
Parrot
Spider
Goldfish

Scrolling menus with multiple

```
<form>
  <label for="pet-select">Choose a pet:</label>

  <select id="pet-select" name="pet-select" size="6" multiple>
    <option value="dog">Dog</option>
    <option value="cat">Cat</option>
    <option value="hamster">Hamster</option>
    <option value="parrot">Parrot</option>
    <option value="spider">Spider</option>
    <option value="goldfish">Goldfish</option>
  </select>
</form>
```



Hidden controls

```
<input type = "hidden" name="variable" value = "val">
```

- The hidden control is used to send information to the form processing application that does not come from the user.
- The hidden control is not visible when the form is displayed in a browser. The hidden data is sent when the form is submitted.

```
<input type = "hidden" name = "pagename" value = "10">
```


Submit and reset buttons

```
<input type = "submit">
```

Submits the form data to the server

```
<input type = "reset">
```

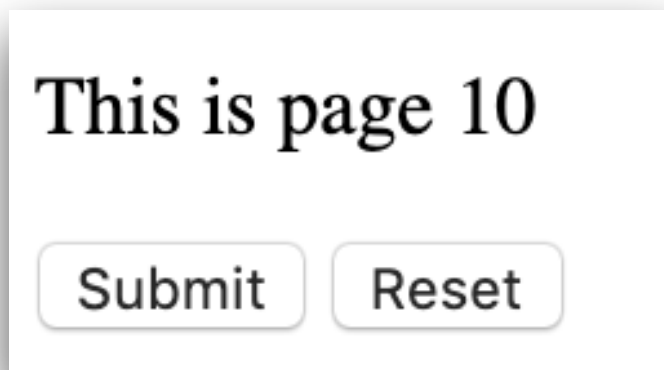
Resets the form controls to their default settings

- Both `submit` and `reset buttons` have specific functions that do not include the entry of the data.
- They are the only form control elements that `do not require` the `name` attribute.
- `By default`, the submit button displays with the label "`Submit`" or "`Submit Query`" and the reset button is labeled "`Reset`".
- Change the text on the button using the `value` attribute.

```
<input type = "submit" value = "Save">
```


hidden controls

```
<form>
  <p>This is page 10</p>
  <input type = "hidden" name = "pagename" value = "10" />
  <input type = "submit" name = "submit" value = "Submit" />
  <input type = "reset" name = "reset" value = "Reset" />
</form>
```



This is page 10

Submit Reset

Demo

Fieldset and legend

- The `<fieldset>` element indicates a logical group of form controls.
- A `<fieldset>` may also include a `<legend>` element that provides a caption for the enclosed fields.
- It draws a box around the grouped elements by default and is crucial for accessibility, especially for users of aural browser.
- `<fieldset disabled>`
 - The attribute `disabled` is set, all form controls are descendants of the `<fieldset>`, are disabled, meaning they are not edible and won't be submitted along with the `<form>`.

<fieldset> element

```
<form>
  <fieldset>
    <legend>Choose your favorite monster</legend>

    <input type="radio" id="kraken" name="monster">
    <label for="kraken">Kraken</label><br/>

    <input type="radio" id="sasquatch" name="monster">
    <label for="sasquatch">Sasquatch</label><br/>

    <input type="radio" id="mothman" name="monster">
    <label for="mothman">Mothman</label>
  </fieldset>
</form>
```

Choose your favorite monster

☐ Kraken
☐ Sasquatch
☐ Mothman

Demo

References

- https://www.tutorialspoint.com/html/html_forms.htm
- https://www.w3schools.com/html/html_forms.asp
- https://www.w3schools.com/html/html_form_elements.asp